MATH 2111

Tutorial 4: Eigenvalues, Eigenspaces and Similarity

WANG Yixiao

Department of Mathematics

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Eigenvectors and Eigenspaces

Definitions

Let A be an $n \times n$ matrix. A nonzero vector $\boldsymbol{v}$ is an eigenvector of A corresponding to λ if

$$A\boldsymbol{v} = \lambda\boldsymbol{v}.$$

The scalar λ is an eigenvalue of A. The eigenspace corresponding to λ is

$$E_\lambda = \mathrm{Nul}(A - \lambda I).$$

Thus λ is an eigenvalue precisely when $(A - \lambda I)\boldsymbol{x} = \boldsymbol{0}$ has a nontrivial solution.

Two useful facts

Eigenvectors corresponding to distinct eigenvalues are linearly independent.

If A is triangular, its eigenvalues are its diagonal entries.

Powers of an eigenvector

If $A\boldsymbol{v} = \lambda\boldsymbol{v}$, then for every integer $k \geq 0$,

$$A^k\boldsymbol{v} = \lambda^k\boldsymbol{v}.$$

This follows by repeated application of A.

Characteristic Equation and Determinants

For an $n \times n$ matrix A, the characteristic polynomial and characteristic equation are

$$p_A(\lambda) = \det(A - \lambda I), \qquad \det(A - \lambda I) = 0.$$

The eigenvalues are the roots of this equation, counted with their algebraic multiplicities.

Determinant facts

$$\det(AB) = \det(A)\det(B), \qquad \det(A^T) = \det(A).$$

For triangular A,

$$\det A = \prod_{i=1}^{n} a_{ii}.$$

A row replacement preserves the determinant; a row interchange changes its sign; scaling one row by c multiplies the determinant by c.

Invertibility and cofactors

$$A \text{ is invertible} \iff \det A \neq 0.$$

Equivalently, 0 is not an eigenvalue of A. A cofactor expansion may be performed along any row or any column. If A_{ij} is obtained by deleting row i and column j, then

$$C_{ij} = (-1)^{i+j} \det A_{ij}.$$

Similarity and Discrete Dynamical Systems

Similarity

Two $n \times n$ matrices A and B are similar if there is an invertible matrix P such that

$$B = P^{-1}AP.$$

Similar matrices have the same characteristic polynomial, and hence the same eigenvalues with the same algebraic multiplicities. The converse need not hold. Similarity is different from row equivalence.

Discrete dynamical systems

For the system

$$\boldsymbol{x}_{k+1} = A\boldsymbol{x}_k, \qquad \boldsymbol{x}_0 \text{ given,}$$

we have $\boldsymbol{x}_k = A^k\boldsymbol{x}_0$. If $\boldsymbol{x}_0 = \sum_{j=1}^{r} c_j\boldsymbol{v}_j$, where $A\boldsymbol{v}_j = \lambda_j\boldsymbol{v}_j$, then

$$\boldsymbol{x}_k = \sum_{j=1}^{r} c_j\lambda_j^k\boldsymbol{v}_j.$$

The long-term behavior is determined by the powers λ_j^k.

Problem 1: Eigenvectors and Eigenspaces

Problem 1

Let

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}, \qquad \boldsymbol{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \boldsymbol{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}, \quad \boldsymbol{w} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

(a) Determine which of $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w}$ are eigenvectors of A, and find the corresponding eigenvalues.

(b) Find a basis for each eigenspace corresponding to the eigenvalues found in part (a).

(c) Explain why $\boldsymbol{u}$ and $\boldsymbol{v}$ are linearly independent.

Solution to Problem 1: Identifying Eigenvectors

Direct multiplication gives

$$A\boldsymbol{u} = \begin{bmatrix} 8 \\ 4 \end{bmatrix} = 4\boldsymbol{u}, \qquad A\boldsymbol{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \boldsymbol{v}.$$

Thus $\boldsymbol{u}$ and $\boldsymbol{v}$ are eigenvectors corresponding to $\lambda = 4$ and $\lambda = 1$, respectively. On the other hand,

$$A\boldsymbol{w} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

is not a scalar multiple of $\boldsymbol{w}$, so $\boldsymbol{w}$ is not an eigenvector.

Solution to Problem 1: Eigenspaces

For $\lambda = 4$,

$$(A - 4I)\boldsymbol{x} = \mathbf{0} \iff -x_1 + 2x_2 = 0,$$

so

$$E_4 = \text{Span}\left\{\begin{bmatrix}2\\1\end{bmatrix}\right\}.$$

For $\lambda = 1$,

$$(A - I)\boldsymbol{x} = \mathbf{0} \iff x_1 + x_2 = 0,$$

so

$$E_1 = \text{Span}\left\{\begin{bmatrix}1\\-1\end{bmatrix}\right\}.$$

Since $4 \neq 1$, eigenvectors $\boldsymbol{u}$ and $\boldsymbol{v}$ are linearly independent.

Problem 2: Repeated Eigenvalues

Problem 2

Let

$$T = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

(a) Find the characteristic polynomial and all eigenvalues, including their algebraic multiplicities.

(b) Find a basis for each eigenspace.

(c) Compare the multiplicity of $\lambda = 3$ as a root of the characteristic polynomial with the dimension of its eigenspace.

Solution to Problem 2: Characteristic Polynomial

Since T is upper triangular, its eigenvalues are its diagonal entries. Thus

$$\begin{aligned} p_T(\lambda) &= \det(T - \lambda I) \\ &= (3-\lambda)^2(1-\lambda). \end{aligned}$$

The eigenvalues are

$$\lambda = 3 \quad \text{(algebraic multiplicity } 2\text{)}, \qquad \lambda = 1 \quad \text{(algebraic multiplicity } 1\text{)}.$$

Solution to Problem 2: Eigenspaces

For $\lambda = 3$,

$$T - 3I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -2 \end{bmatrix},$$

so $x_2 = x_3 = 0$ and

$$E_3 = \text{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \right\}.$$

For $\lambda = 1$,

$$T - I = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

so $x_1 = x_2 = 0$ and

$$E_1 = \text{Span}\left\{ \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Although 3 is a repeated root, its eigenspace is only one-dimensional.

Problem 3: Characteristic Equation

Problem 3

Let

$$S = \begin{bmatrix} 4 & 0 & -1 \\ 0 & 3 & 0 \\ 3 & 0 & 0 \end{bmatrix}.$$

(a) Find the characteristic polynomial and all eigenvalues, using a cofactor expansion along the second row of $S - \lambda I$.

(b) Find a basis for the eigenspace corresponding to $\lambda = 3$.

(c) Determine whether S is invertible, and show that S^T has the same characteristic polynomial as S.

Solution to Problem 3: Characteristic Polynomial

We have

$$S - \lambda I = \begin{bmatrix} 4-\lambda & 0 & -1 \\ 0 & 3-\lambda & 0 \\ 3 & 0 & -\lambda \end{bmatrix}.$$

Expanding along the second row,

$$\begin{aligned} \det(S - \lambda I) &= (3-\lambda) \begin{vmatrix} 4-\lambda & -1 \\ 3 & -\lambda \end{vmatrix} \\ &= (3-\lambda)\big((4-\lambda)(-\lambda) + 3\big) \\ &= (3-\lambda)(\lambda^2 - 4\lambda + 3) \\ &= (3-\lambda)^2(1-\lambda). \end{aligned}$$

Thus the eigenvalues are 3 (algebraic multiplicity 2) and 1 (algebraic multiplicity 1).

Solution to Problem 3: Eigenspace and Determinant

For $\lambda = 3$,

$$S - 3I = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 3 & 0 & -3 \end{bmatrix}.$$

The equation $(S - 3I)\boldsymbol{x} = \mathbf{0}$ gives $x_1 = x_3$, with x_2 free. Hence

$$E_3 = \text{Span}\left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\}.$$

Also,

$$\det S = \det(S - 0I) = 3^2 \cdot 1 = 9 \neq 0,$$

so S is invertible and 0 is not an eigenvalue.
The transpose property of the determinant gives

$$\begin{aligned} \det(S^T - \lambda I) &= \det\big((S - \lambda I)^T\big) \\ &= \det(S - \lambda I). \end{aligned}$$

Thus S^T and S have the same characteristic polynomial.

Problem 4: Similarity and Its Limitations

Problem 4

Let

$$D = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}, \qquad P = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

(a) Compute $B = P^{-1}DP$ and verify that B and D have the same characteristic polynomial.

(b) The matrices T and S in Problems 2 and 3 have the same characteristic polynomial. Are they necessarily similar? Justify your answer.

(c) Are T and S row equivalent? Explain the difference between this conclusion and similarity.

Solution to Problem 4: A Similarity Transformation

The inverse of P is

$$P^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}.$$

Using this inverse,

$$B = P^{-1}DP = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$$
$$= \boxed{\begin{bmatrix} 0 & -4 \\ 2 & 6 \end{bmatrix}}.$$

Its characteristic polynomial is

$$\begin{aligned} \det(B - \lambda I) &= \begin{vmatrix} -\lambda & -4 \\ 2 & 6 - \lambda \end{vmatrix} \\ &= \lambda^2 - 6\lambda + 8 \\ &= (2 - \lambda)(4 - \lambda). \end{aligned}$$

This is also $\det(D - \lambda I)$, as expected for similar matrices.

Solution to Problem 4: Same Eigenvalues Do Not Suffice

Problems 2 and 3 give

$$\det(T - \lambda I) = \det(S - \lambda I) = (3 - \lambda)^2(1 - \lambda).$$

However,

$$\dim E_3(T) = 1, \qquad \dim E_3(S) = 2.$$

If $S = Q^{-1}TQ$ for an invertible Q, then

$$S - 3I = Q^{-1}(T - 3I)Q.$$

Consequently, $\boldsymbol{x} \in E_3(S)$ if and only if $Q\boldsymbol{x} \in E_3(T)$. Since Q is invertible, it gives a one-to-one correspondence between these eigenspaces, so their dimensions must be equal. This contradiction shows that T and S are not similar.

Both T and S are invertible (their determinants are 9), so each is row equivalent to I_3. Hence they are row equivalent to each other, even though they are not similar.

Problem 5: Discrete Dynamical Systems

Problem 5

Consider the discrete dynamical system

$$\boldsymbol{x}_{k+1} = A\boldsymbol{x}_k, \qquad A = \begin{bmatrix} \frac{1}{4} & \frac{3}{4} \\ \frac{3}{4} & \frac{1}{4} \end{bmatrix}, \qquad \boldsymbol{x}_0 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

(a) Find the eigenvalues of A and a corresponding eigenvector for each eigenvalue.

(b) Express $\boldsymbol{x}_0$ as a linear combination of these eigenvectors, and derive a formula for $\boldsymbol{x}_k$.

(c) Find $\lim_{k\to\infty} \boldsymbol{x}_k$. Does the sequence approach its limit from one side or alternate around it?

Solution to Problem 5: Eigenvector Expansion

The characteristic polynomial is

$$\begin{aligned}\det(A-\lambda I) &= \left(\frac{1}{4}-\lambda\right)^2 - \frac{9}{16} \\ &= (1-\lambda)\left(-\frac{1}{2}-\lambda\right).\end{aligned}$$

Thus the eigenvalues are 1 and $-\frac{1}{2}$. Corresponding eigenvectors are

$$\boldsymbol{u}_1 = \begin{bmatrix}1\\1\end{bmatrix}, \qquad \boldsymbol{u}_2 = \begin{bmatrix}1\\-1\end{bmatrix}.$$

They are linearly independent because their eigenvalues are distinct. The initial vector has the decomposition

$$\boldsymbol{x}_0 = \begin{bmatrix}1\\0\end{bmatrix} = \frac{1}{2}\boldsymbol{u}_1 + \frac{1}{2}\boldsymbol{u}_2.$$

Using $A^k\boldsymbol{u}_i = \lambda_i^k \boldsymbol{u}_i$, we obtain

$$\boldsymbol{x}_k = \frac{1}{2}\boldsymbol{u}_1 + \frac{1}{2}\left(-\frac{1}{2}\right)^k \boldsymbol{u}_2.$$

Solution to Problem 5: Long-Term Behavior

Substituting the eigenvectors gives

$$\boxed{\boldsymbol{x}_k = \frac{1}{2}\begin{bmatrix}1\\1\end{bmatrix} + \frac{1}{2}\left(-\frac{1}{2}\right)^k \begin{bmatrix}1\\-1\end{bmatrix}.}$$

Equivalently,

$$\boldsymbol{x}_k = \begin{bmatrix}\frac{1}{2} + \frac{1}{2}(-\frac{1}{2})^k \\ \frac{1}{2} - \frac{1}{2}(-\frac{1}{2})^k\end{bmatrix}.$$

Since $\left|-\frac{1}{2}\right| < 1$,

$$\boxed{\lim_{k\to\infty} \boldsymbol{x}_k = \begin{bmatrix}\frac{1}{2}\\ \frac{1}{2}\end{bmatrix}.}$$

The factor $(-\frac{1}{2})^k$ changes sign at each step while its magnitude decreases. Therefore the sequence alternates around its limiting state, with the distance to the limit tending to zero.